System and Device Programming

Review Exercises

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License Information

Exercise 01 (C and UNIX IO)

Please, recall the SDP standards
C, POSIX, C++

- An ASCII file stores a sequence of records
  - Each record includes 3 fields
    - An integer value, a string, and a real number
  - All three fields have variable size and are separated by a variable number of white spaces

```
15345 Acceptable 26.50
146467 Average 23.75
. . .
```

Exercise 01 (C and UNIX IO)

- Write a segment of C code that stores the records into a **list of structures** of type record_t, using
  - The C library
  - The the POSIX/UNIX library

```
15345 Acceptable 26.50
146467 Average 23.75
. . .
```

```
#define N 100

typedef struct record_s {
  int i;
  char s[N];
  float f;
  struct record_s *next;
} record_t;
```

- Observation
  - Keep in mind that the UNIX system call **read** manipulates files in binary form

Solution 1

C Library

Simple solution:
The file has a strict and regular format

```
#define L 2000

fp = fopen (..., "r");
head = NULL;
while (fgets (line, L, fp) != NULL) {
  r = (record_t *) malloc (1 * sizeof (record_t));
  if (r==NULL) { ... error ... }
  sscanf (line, "%ld %s %f", &r->n, r->s, &r->f);
  r->next = head; head = r;
}
fclose (fd);
```

We can also use
`fscanf(line,"%ld %s %f",...)!=EOF`

Solution 2

POSIX Library

Complex solution:
read manipulates binary files

```
typedef struct tmp_s {
  int i;
  char s[N];
  float f;
} tmp_t;

tmp_t r;

fd = open (..., O_RDONLY);
head = NULL;
while (read (fd, &r, sizeof (tmp_t)) != 0) {
  ...
}
close (fd);
```

Buggy

Records do not have a fixed length
(sizeof (tmp_t))

Solution 3

POSIX Library

```
fd = open (..., O_RDONLY);
head = NULL;

close (fd);
```

Read entire line and convert

```
do {
  i = 0;
  do {
    not_end = read (fd, &c, sizeof(char));
    if (not_end != 0) line[i++] = c;
  } while (c!='\n' && not_end);
  if (not_end) {
    r = (record_t *) malloc (1 * sizeof (record_t));
    sscanf (line, "%ld %s %f", &r->n, r->s, &r->f);
    r->next = head; head = r;
  }
} while (not_end);
```

Solution 4

POSIX Library

Read and convert each field (long int, string, float)

```
do {
  n = 0;
  do {
    not_end = read (fd, &c, sizeof(char));
    if (c!=' ' && not_end) n = n * 10 + ((int) (c-'0'));
  } while (c!=' ' && not_end);
  c = skip_spaces (fd);
  i = 0; tmp1[i++] = c;
  do {
    not_end = read (fd, &c, sizeof(char));
    if (c!=' ' && not_end) tmp1[i++] = c;
  } while (c!=' ' && not_end);
  tmp1[i] = '\0';
```

Solution 4

```
  c = skip_spaces (fd);
  i = 0;
  tmp2[i++] = c;
  do {
    not_end = read (fd, &c, sizeof(char));
    if (c!=' ' && not_end) tmp2[i++] = c;
  } while (c!='\n' && not_end);
  if (not_end) {
    r = (record_t *) malloc (1 * sizeof (record_t));
    r->n = n; strcpy (r->s, tmp1);  r->f = atof (tmp2);
    r->next = head;
    head = r;
  }
} while (not_end);
```

Exercise 02 (fork)

- Draw the control flow graph and the process generation tree of the following C code snippet
- Indicate the generated output

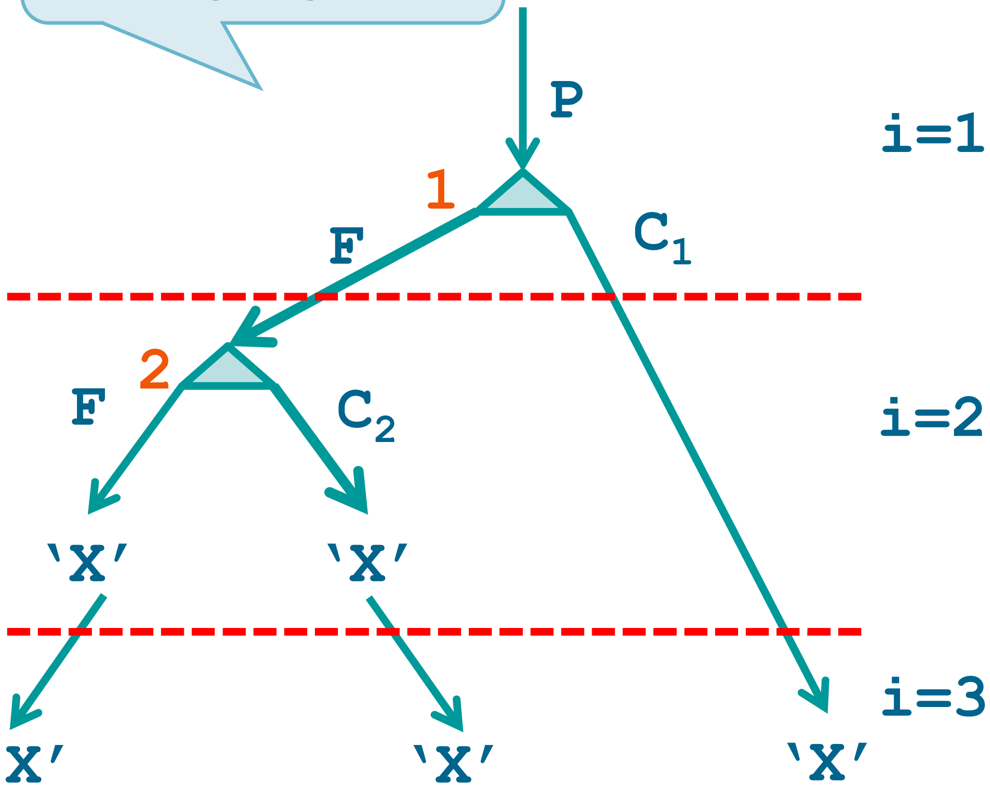
```
int main () {
  int i, pid;
  setbuf(stdout,0);
  for (i=1; i<3; i++) {
    pid = fork ();
    i++;
    if (pid!=0) {
      fork();
      printf("X");
    }
  }
  printf("X");
  return 1;
}
```

Solution

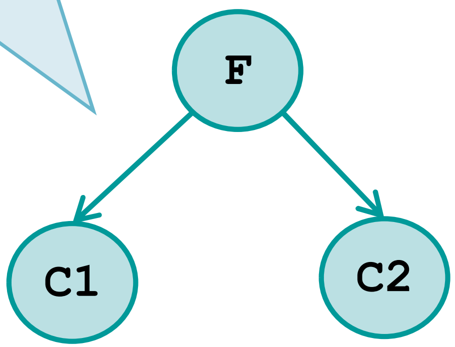
```
for (i=1; i<3; i++) {
  pid = fork ();   // 1
  i++;
  if (pid!=0) {
    fork();
    printf("X");   // 2
  }
}
printf("X");
```

Output: XXXXX

Control Flow Graph (CFG)



Process Generation Tree



Exercise 03 (fork, exec, system)

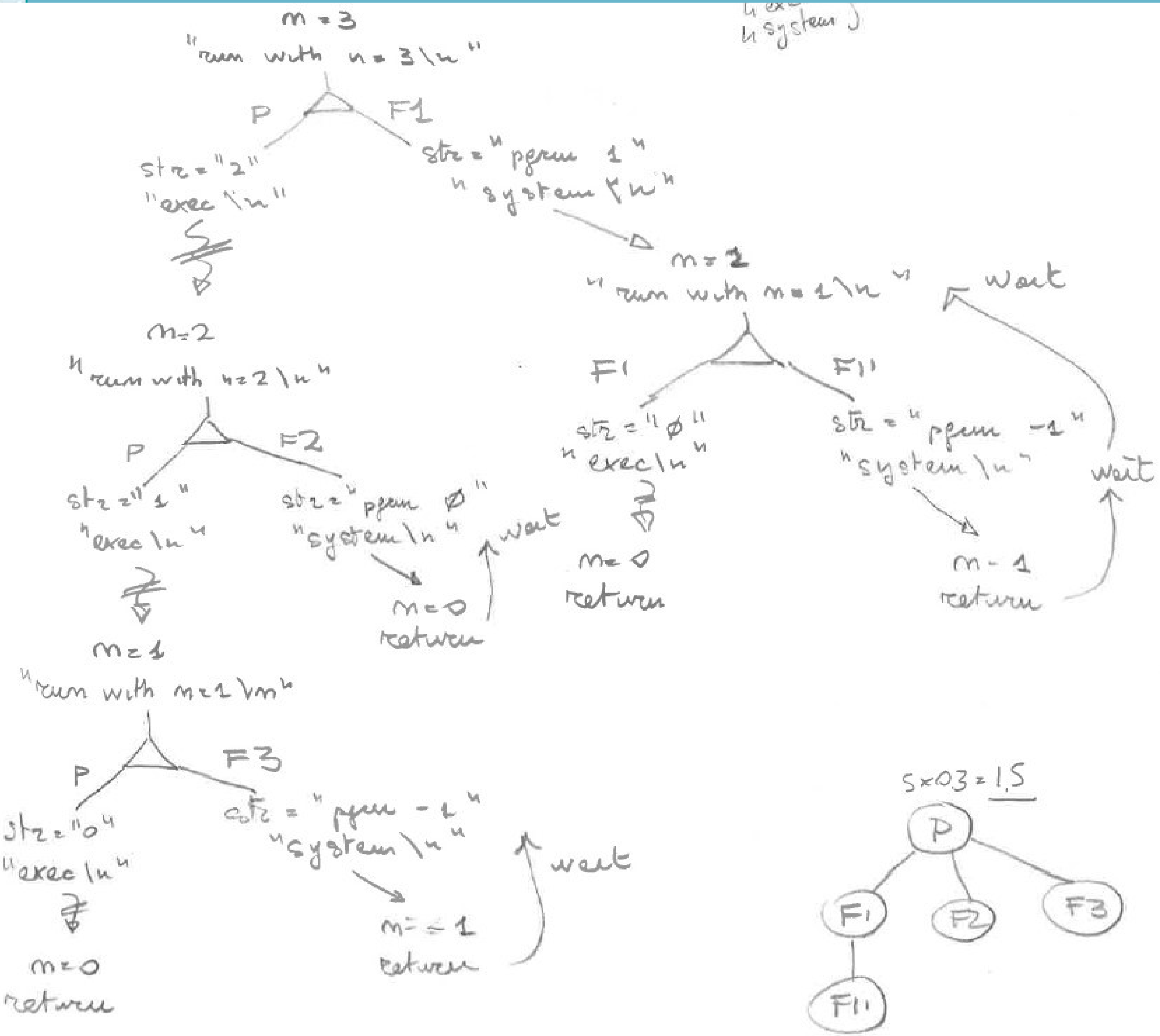
- Assume the following program runs with a unique parameter on the command line, i.e., the integer value 3
- Draw
  - The control flow graph
  - The process generation tree
- Indicate what it displays on standard output

Exercise 03 (fork, exec, system)

```
int main (int argc, char *argv[]) {
  char str[20];
  int n = atoi (argv[1]);
  setbuf (stdout, 0);
  if (n<=0) return (1);
  printf ("run with n=%d\n", n);
  if (fork()>0) {
    sprintf (str, "%d", n-1);
    printf ("exec\n");
    execlp (argv[0], argv[0], str, (char *) 0);
  } else {
    sprintf (str, "%s %d", argv[0], n-2);
    printf ("system\n");
    system (str);
  }
  wait ((int *) 0);
   return (1);
}
```

```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>
```

Solution



n = 3
"run with n = 3\n"
P
F1
str = "2"
"exec \n"
str = "pgrm 1"
"system \n"
n = 1
"run with n = 1\n"
wait
n = 2
"run with n = 2\n"
F1
F11
P
F2
str = "∅"
"exec\n"
str = "pgrm -1"
"system \n"
wait
str = "1"
"exec \n"
str = "pgrm ∅"
"system \n"
wait
n = 0
return
n = -1
return
n = 0
return
n = 1
"run with n = 1\n"
P
F3
str = "0"
"exec \n"
str = "pgrm -1"
"system \n"
wait
n = -1
return
n = 0
return
5 × 0.3 = 1.5
P
F1
F2
F3
F11

Solution

- Output
  - run with n=3 1 time
  - run with n=2 1 time
  - run with n=1 2 times
  - exec 4 times
  - system 4 times

Exercise 04 (orphans and zombies)

- Illustrate the following concepts
  - Orphan process
  - Zombie process
- Report two code segments
  - The first one to generate an orphan process
  - The second one to generate a zombie process

Solution

- A process
  - Is said to be an **orphan** process if its parent ends before waiting for it
    - When the parent ends, the orphan process is "adopted" by a special process, typically the init process
  - Is said to be a **zombie** process if it terminates before its parent issuer a wait
    - Its Process Control Bock (PCB) is stored by the OS to be avalable to the parent when the parent ends
    - It wastes OS resources

Solution

```
if(fork()){
    exit(0);
} else{
    // This process waits for a second
    // Thus, the parent should terminate before it
    sleep(1);
    exit(0);
}
```

Parent process

Orphan process

```
if(fork()) {
  // This process waits for a second
  // Thus, the child should terminate before it
  sleep(1);
  exit(0);
} else {
  exit(0);
}
```

Parent process

Zombie process

Solution

- An init and a zombie process on a Linux machine

```
USER   PID %CPU %MEM    VSZ   RSS TTY    STAT START   TIME COMMAND
root     1  0.3  0.0 166876 12112 ?      Ss   22:56   0:01 /sbin/init splash
root     2  0.0  0.0      0     0 ?      S    22:56   0:00 [kthreadd]
root     3  0.0  0.0      0     0 ?      I<   22:56   0:00 [rcu_gp]
root     4  0.0  0.0      0     0 ?      I<   22:56   0:00 [rcu_par_gp]
...
quer  2274  0.0  0.0  11540  5840 pts/0  Ss   22:57   0:00 bash
quer  3065  2.1  0.1 770444 74016 pts/0  Sl   23:00   0:03 emacs
quer  3082  0.0  0.0   2772   948 pts/0  S    23:00   0:00 ./a
quer  3084  0.0  0.0      0     0 pts/0  Z    23:00   0:00 [a] <defunct>
quer  3119  0.0  0.0  12992  3784 pts/0  R+   23:02   0:00 ps -aux
```

Exercise 05 (signals)

- A process P
  - Generates two child processes P1 and P2
  - It goes into a wait state and
    - Every time it receives a message from P1 (P2), it displays the message "Signal received from P1(P2)"
    - If it receives 3 signals from the same process, it terminates processes P1 and P2 and it ends
- Process P1 and P2 run through an infinite cycle
  - They wait for a random time and then send a signal to the parent
    - P1 sends signals SIGUSR1
    - P2 sends signals SIGUSR2

Solution

```
#include <signal.h>
#include <sys/types.h>
#include <sys/wait.h>
#include <unistd.h>
#include <stdio.h>
#include <stdlib.h>
```

```
int last_sig = -1;
int last_last_sig = -1;
int finish = 0;


void sign_handler(int sig){
  if (sig==SIGUSR1)
    printf("Signal received from P1\n");
  else if (sig==SIGUSR2)
    printf("Signal received from P2\n");
  if (sig == last_sig && last_sig == last_last_sig) {
    finish = 1;
  } else {
    last_last_sig = last_sig;
    last_sig = sig;
  }
}
```

Solution

```
int main() {
  int pid1, pid2;
  char cmd[100];

  if ( (signal(SIGUSR1, sign_handler) == SIG_ERR)
    || (signal(SIGUSR2, sign_handler) == SIG_ERR) ) {
    printf("Error initializing signal handler");
    exit(-1);
  }
  pid1 = fork();
  if (pid1==0) {
    while (1) {
      sleep( rand()%2 );
      kill(getppid(), SIGUSR1);
    }
  } else {
```

Child 1
P1

Solution

```
    pid2 = fork();
    if (pid2==0) {
      while (1) {
        sleep(rand()%3);
        kill(getppid(), SIGUSR2);
      }
    } else {
      while (1) {
        pause();
        if (finish) {
          sprintf(cmd, "kill -9 %d", pid1); system(cmd);
          sprintf(cmd, "kill -9 %d", pid2); system(cmd);
          exit(0);
        }
      }
    }
  return (0);
}
```

Child 2
P2

Parent
P

Kill children
before ending

Exercise 06 (pipes)

- A program runs three processes (P1, P2, and P3) connected to each other by three pipes (P1-P2, P2-P3, and P3-P1)
- The three processes work in a circular manner
  - Each process Pi (i.e. P1, P2, or P3) reads an integer from standard input
  - Transfer the integer to the next process (i.e. P2, P3, or P1, respectively)
  - The next process in the chain sleeps a number of seconds equal to that value and it repeats the same operation
- Assume that processes do not terminate

Solution

```
#include <stdio.h>
#include <string.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>

static void p1(int, int);
static void p2(int, int);
static void p3(int, int);
```

Solution

If the pipe is closed at the other end:
Read returns 0
Write returns SIGPIPE

```
int main () {
  int p12[2], p23[2], p31[2];
  setbuf (stdout, 0);
  pipe (p12); pipe (p23); pipe (p31);
  if (fork()!=0) {
    close(p12[0]); close(p23[0]); close(p23[1]); close(p31[1]);
    p1 (p31[0], p12[1]);
  } else {
    if (fork()!=0) {
      close(p12[1]); close(p23[0]); close(p31[0]); close(p31[1]);
      p2 (p12[0], p23[1]);
    } else {
      close(p12[0]); close(p12[1]); close(p23[1]); close(p31[0]);
      p3 (p23[0], p31[1]);
    }
  }
  return (0);
}
```

Solution

```
static void p1 (int pr, int pw) {
  int n;
  n = 0;
  while (1) {
    fprintf (stdout, "P1 waiting: %d secs\n", n);
    sleep (n);
    fprintf (stdout, "P1 reading: ");
    scanf ("%d", &n);
    write (pw, &n, sizeof (int));
    read (pr, &n, sizeof (int));
  }
  return;
}
```

One process must write first to start the pipeline

Solution

```
static void p2 (int pr, int pw) {
  int n;
  while (1) {
    read (pr, &n, sizeof (int));
    fprintf (stdout, "P2 waiting: %d secs\n", n);
    sleep (n);
    fprintf (stdout, "P2 reading: ");
    scanf ("%d", &n);
    write (pw, &n, sizeof (int));
  }
  return;
}
```

The other processes must read first

Solution

```
static void p3 (int pr, int pw) {
  int n;
  while (1) {
    read (pr, &n, sizeof (int));
    fprintf (stdout, "P3 waiting: %d secs\n", n);
    sleep (n);
    fprintf (stdout, "P3 reading: ");
    scanf ("%d", &n);
    write (pw, &n, sizeof (int));
  }
  return;
}
```

Equivalent to p2
(can share ther same function)

The other processes must read first

Exercise 07 (IO and pipes)

- A file stores an undefined number of strings, each one stored on a different file line
- Two processes P1 and P2 want to
  - Use the file to exchange strings between them
  - Synchronize their R/W operations on the file using one or more pipes
    - When P1 runs P2 waits and vice versa

Exercise 07 (IO and pipes)

- Each process
  - Reads the contents of the file and displays it on standard output
  - Store a new set of strings into the file
    - The new set of strings is read from the keyboard
    - The string end stop this phase
- The entire process must terminate when either P1 or P2 reads the string END

Solution

```
... includes ...
#define N 100
... prototypes ...

int main (int argc, char **argv) {
  int fd1[2], fd2[2];
  int childPid;
  pipe (fd1);
  pipe (fd2);
  childPid = fork();
  if (childPid == 0) {
    child (argv[1], fd1, fd2);
  } else {
    parent (argv[1], fd1, fd2);
  }
  wait ((void *)0);
  exit (1);
}
```

See source code (in u02s02e) for all implementation details

Solution

```
void  parent (char *name, int *fd1, int *fd2) {
  char c;
  int stop = 0;
  close (fd1[0]); close (fd2[1]);
  while (stop==0) {
    printf ("PARENT WAKE UP (PID=%d) \n", getpid());
    read_file (name);
    stop = write_file (name);
    c = (stop==0) ? '0' : '1';
    write (fd1[1], &c, sizeof (char));
    if (stop==1)
      break;
    read (fd2[0], &c, sizeof (char));
    stop = (c=='0') ? 0 : 1;
  }
  close (fd1[1]); close (fd2[0]);
  return;
}
```

Solution

```
void child (char *name, int *fd1, int *fd2) {
  char c;
  int stop = 0;
  close (fd1[1]); close (fd2[0]);
  while (stop==0) {
    read (fd1[0], &c, sizeof (char));
    printf ("CHILD WAKE UP (PID=%d) \n", getpid());
    if (c=='1')
      break;
    read_file (name);
    stop = write_file (name);
    c = (stop==0) ? '0' : '1';
    write (fd2[1], &c, sizeof (char));
  }
  close (fd1[0]); close (fd2[1]);
  return;
}
```

Solution

```
void read_file (char *name) {
  FILE *fp;
  char str[N];

  fp = fopen (name, "r");
  while (fscanf (fp, "%s", str) != EOF) {
    fprintf (stdout, "   %s\n", str);
  }
  fclose (fp);

  return;
}
```

Solution

```
int write_file (char *name) {
  FILE *fp;
  char str[N];
  fp = fopen (name, "w");
  do {
    fprintf (stdout, "   str: ");
    scanf ("%s", str);
    if (strcmp (str, "end") != 0 && strcmp (str, "END") != 0) {
      fprintf (fp, "   %s\n", str);
    }
  } while (strcmp (str, "end") != 0 && strcmp (str, "END") != 0);
  fclose (fp);
  if (strcmp (str, "END") == 0)
    return 1;
  else
    return 0;
}
```

Exercise 08 (thread analysis)

- Suppose that the following program is executed passing the value 5 in the command line
- Report the exact output it generates

```
... includes ...

... prototypes ...

int main (int argc, char *argv[]) {
  pthread_t thread;
  int n = atoi (argv[1]);
  setbuf (stdout, 0);
  pthread_create (&thread, NULL, T1, &n);
  pthread_join (thread, NULL);
  return 1;
}
```

See source code (in u02s02e) for all implementation details

Exercise

```
void *T1 (void *p){
  pthread_t thread;
  int *pn = (int *) p;
  int n = *pn;
  if (n>0) {
    n-=2;
    printf ("(%d)", n);
    pthread_create (&thread, NULL, T2, &n);
  }
  pthread_join (thread, NULL);
  pthread_exit (NULL);
}
```

```
void *T2 (void *p){
  pthread_t thread;
  int *pn = (int *) p;
  int n = *pn;
  if (n>0) {
    n+=1;
    printf ("[%d]", n);
    pthread_create (&thread, NULL, T1, &n);
  }
  pthread_join (thread, NULL);
  pthread_exit (NULL);
}
```

Exercise

```
void *T1 (void *p){
  pthread_t thread;
  int *pn = (int *) p;
  int n = *pn;
  if (n>0) {
    n-=2;
    printf ("(%d)", n);
    pthread_create (&thread, NULL, T2, &n);
  }
  pthread_join (thread, NULL);
  pthread_exit (NULL);
}
```

```
void *T2 (void *p){
  pthread_t thread;
  int *pn = (int *) p;
  int n = *pn;
  if (n>0) {
    n+=1;
    printf ("[%d]", n);
    pthread_create (&thread, NULL, T1, &n);
  }
  pthread_join (thread, NULL);
  pthread_exit (NULL);
}
```

Output

```
(3)[4](2)[3](1)[2](0)
```

Exercise 09 (synchronization)

- A program executes a high number of threads T
- In order not to exceed the hardware resources, it forces the code section SC to be executed simultaneously by a maximum of N threads, with N << T
  - This behavior is achieved by defining a semaphore sem initialized to the value N, and then accessing SC only through the following code section

```
sem_wait (sem);
SC
sem_post (sem);
```

Exercise 09 (synchronization)

- Later on, the programmer wants to run some other threads that consume twice the hardware resources of the former threads
  - The programmer's idea is to manage the access to SC of these threads through the following piece of code

```
sem_wait (sem);
sem_wait (sem);
SC
sem_post (sem);
sem_post (sem);
```

- Is this correct?

Solution

- The solution is prone to deadlock
  - If N threads execute one sem_wait, before the second sem_wait, is executed by at least one of them, they would be blocked because no thread could enter the critical section (CS), and consequently execute the sem_post, which allows other threads to enter the critical section
- A possible solution is to make the couple of sem_wait "atomic"
  - We can use mutual exclusion on sem_wait
  - Pay attention to single sem_wait to

Solution

```
pthread_muted_t m;

...

pthread_mutex_init (&m, NULL);

...

ptrhead_mutex_lock (&m);
sem_wait (sem);
sem_wait (sem);
ptrhead_mutex_unlock (&m);
SC
sem_post (sem);
sem_post (sem);
```

Exercise 10 (synch - pseudo-code)

- In a system there are **two** threads
  - One of type A and one of type B
  - Thread A can call function **fA**
  - Thread B can call function **fB**
- The
  - First thread to call **fA** or **fB** is blocked
  - Second thread to call **fA** or **fB** is not blocked and it also unlocks the thread previously blocked
- Functions **fA** and **fB** can be reused by threads A and B an indefinite number of times

Exercise 10 (synch - pseudo-code)

- Using the pseudo-code primitives wait and signal write functions **fA** and **fB**
  - Use the minimum number of semaphores
  - Define all global variables and all semaphore initialization values
- Example
  - A calls the function **fA** and it blocks
  - B calls the function **fB** and it unlocks A
  - B calls the function **fB** and it blocks
  - A calls the function **fA** and unlocks B

Solution

```
init(m, 1);
init(s, 0);
int flag=0;


fA() {
  wait(m);
  if (flag==0) {
    flag=1;
    signal(m);
    wait(s);
  } else {
    flag=0;
    signal(m);
    signal(s);
  }
}
```

If flag==0 no one is waiting
flag==1 someone is waiting

Noone is waiting
Go to sleep

fB is waiting
(free it)

```
fB() {
  wait(m);
  if (flag==0) {
    flag=1;
    signal(m);
    wait(s);
  } else {
    flag=0;
    signal(m);
    signal(s);
  }
}
```

m is the mutual
exclusion mutex

fA is waiting
(free it)

s is the waiting
semaphore

Exercise 11 (synch - C)

- A program includes several threads and a shared variable that indicates one of two possible states (RED or GREEN)
- Each thread can call the
  - Function **change** to modify the state variable
    - If a thread calls change when the state is GREEN (RED) it changes the shared state variable to RED (GREEN)
  - Function **red** (**green**)
    - It blocks a thread when the state is GREEN (RED)
    - Does not block a thread when the state is RED (GREEN)

Exercise 11 (synch - C)

- Implement functions change, red, and green
  - Assume that the initial state is RED for all threads
- Example of execution
  - T1 calls red and it does not block
  - T2 calls red and it does not block
  - T3 calls green and it blocks
  - T4 calls green and it blocks
  - T1 calls red and it does not block
  - T5 calls change, the status changes to GREEN, and T3 and T4 are released (unblocked)
  - T5 calls green and it does not block

Solution

```c
#define RED   0
#define GREEN 1
int state = RED;
sem_t s_red;
sem_t s_green;
sem_t m;
sem_init(&s_red, 0, 1);
sem_init(&s_green, 0, 0);
sem_init(&m, 0, 1);
red(){
  sem_wait(&s_red);
  sem_post(&s_red);
}
green(){
  sem_wait(&s_green);
  sem_post(&s_green);
}
```

```c
change(){
  sem_wait(&m);
  if(state == RED){
    state = GREEN;
    sem_wait(&s_red);
    sem_post(&s_green);
  } else {
    state = RED;
    sem_wait(&s_green);
    sem_post(&s_red);
  }
  sem_post(&m);
}
```

Exercise 12 (synch)

- A program runs N threads of type A and N threads of type B
  - N is a positive integer passed on the command line to the program (e.g., 10)
  - All threads are identified by their category (i.e., A or B) and a creation index (i.e., 1, 2, ..., N)
- We want to coordinate the effort of all threads to always **run** a thread of the category A **before** one thread of the category B

Exercise 12 (synch)

- In the following there are **two** examples of correct executions with N=10

```
A3 B1 A1 B8 A2 B7 A8 B6 A9 B3 A7 B9 A6 B2 A5 B4 A4 B5 A10 B10
A10 B5 A6 B1 A7 B3 A1 B4 A3 B2 A2 B9 A8 B8 A5 B6 A4 B10 A9 B7
```

- Note that the order in which the threads of each category are executed is not fixed but depends on the threads scheduling
- Write the code using the C language
  - Realize the complete program, including the creation of the threads

Solution

```
... includes ...
... prototypes ...
sem s_a, s_b;

int main(int argc, char *argv[]){
  int N ;
  int i = 0;
  pthread_t *threads_A;
  pthread_t *threads_B;
  int *id;

  N = atoi(argv[1]);

  sem_init(&s_a, 0, 1);
  sem_init(&s_b, 0, 0);
  threads_A = (pthread_t*)malloc(N*sizeof(pthread_t));
  threads_B = (pthread_t*)malloc(N*sizeof(pthread_t));
  id = (int*)malloc(N*sizeof(int));
```

Semaphores initialization

Data structure allocation

OS IDs

User IDs

Solution

Thread generation

```
  for (i=0; i<N;i++){
    id[i] = i;
    pthread_create(&threads_A[i], NULL, TA, (void *)&(id[i]));
    pthread_create(&threads_B[i], NULL, TB, (void *)&(id[i]));
  }

  for (i=0; i<N;i++){
    pthread_join(threads_A[i], NULL);
    pthread_join(threads_B[i], NULL);
  }

  free(threads_A);
  free(threads_B);
  free(id);
  return 0;
}
```

Thread wait (join)

Data structure deallocation

Solution

Semaphores initialization (main) !

```
sem_init(&s_a, 0, 1);
sem_init(&s_b, 0, 0);
```

```
static void *TA(void *arg){
  int id;
  id = *((int*)arg);
  sem_wait(&s_a);
  printf("A%d ", id);
  sem_post(&s_b);
  pthread_exit(NULL);
}
```

```
static void *TB(void *arg){
  int id;
  id = *((int*)arg);
  sem_wait(&s_b);
  printf("B%d ", id);
  sem_post(&s_a);
  pthread_exit(NULL);
}
```

Exercise 13 (IO, thread, synch)

- A file, of undefined length and in ASCII format, contains a list of integer numbers
- Write a program that, after receiving a value **k** (integer) and a string from command line, generates **k** threads and wait them
- Each thread
  - Reads the file **concurrently**, and performs the **sum** of the integer numbers it reads from file
  - When the end of file is reached, it **prints** the number of integer numbers it has read and the computed sum
  - Terminates

Exercise 13 (IO, thread, synch)

- After all threads terminate, the main thread has to print the total number of integer numbers and the total sum
- Example

File format: file.txt

```
7
9
2
-4
15
0
3
```

Example of execution

2 Threads

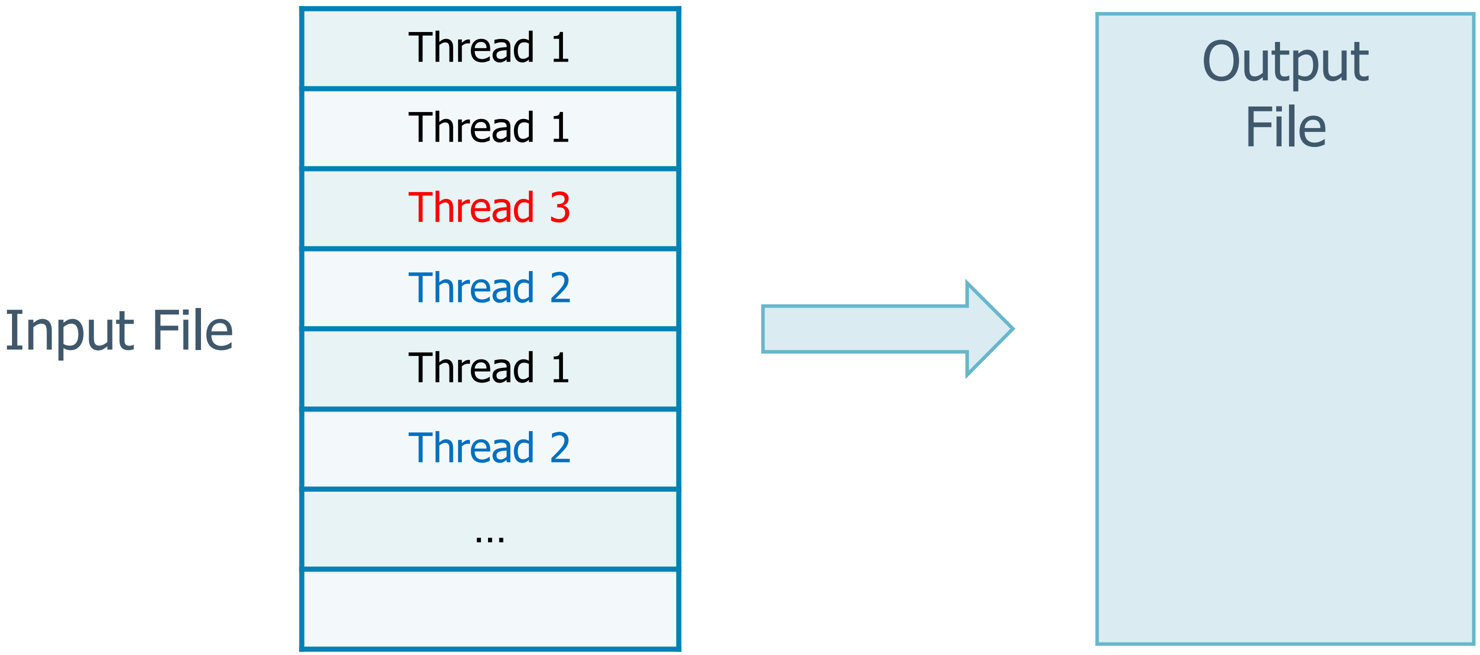
```
➢  pgrm 2 file.txt
Thread 1: Sum=18 #Line=3
Thread 2: Sum=14 #Line=4
Total   : Sum=32 #Line=7
```

Solution

The faster thread gets next record

Solution 1

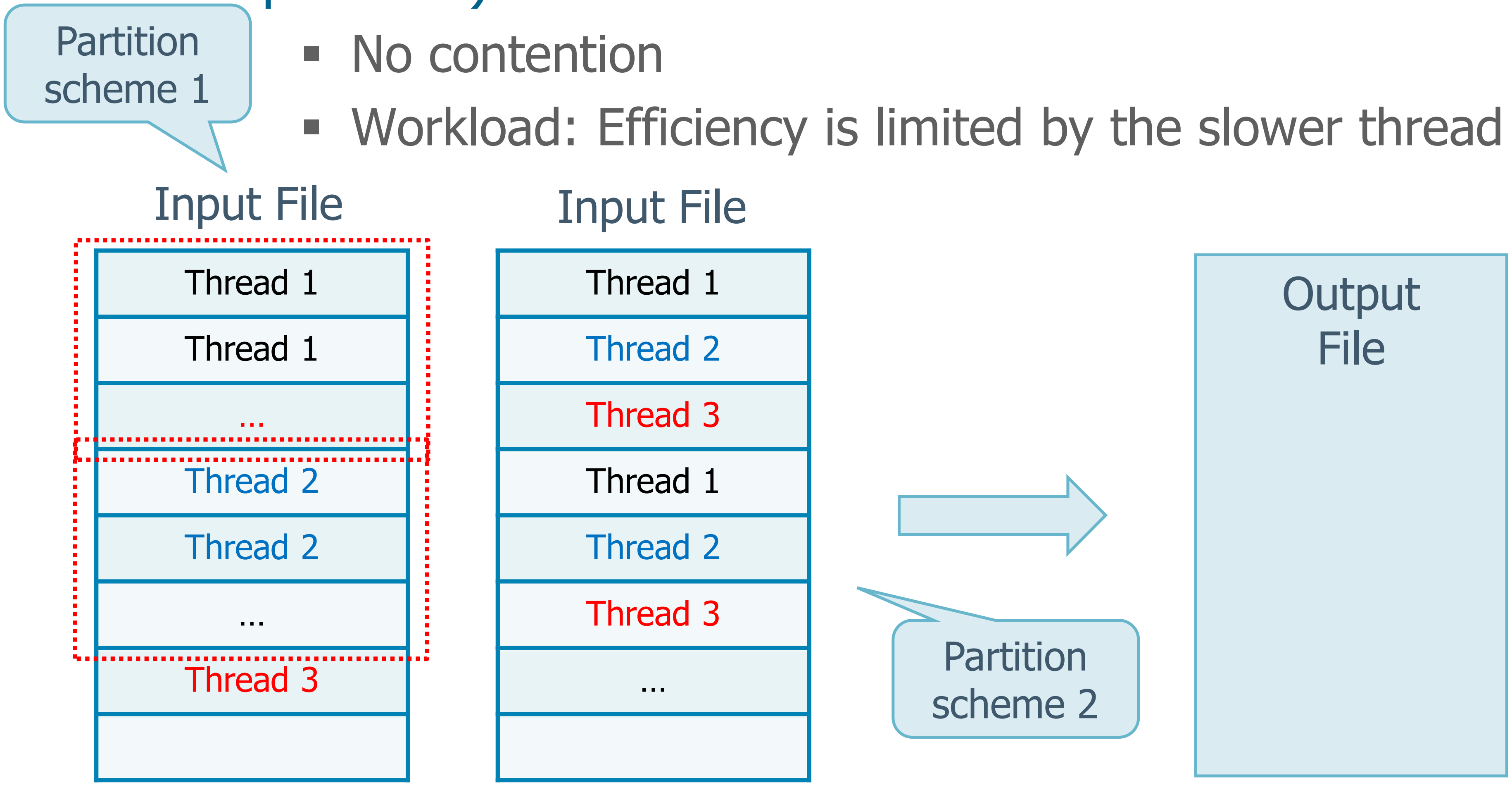
- Let the threads run freely (**dynamic** partition)
  - Simple implementation
    - Just protect file I/O (read/write)
  - **High** thread contention
    - Quite slow in practice



Solution

Each thread gets its own part of the file

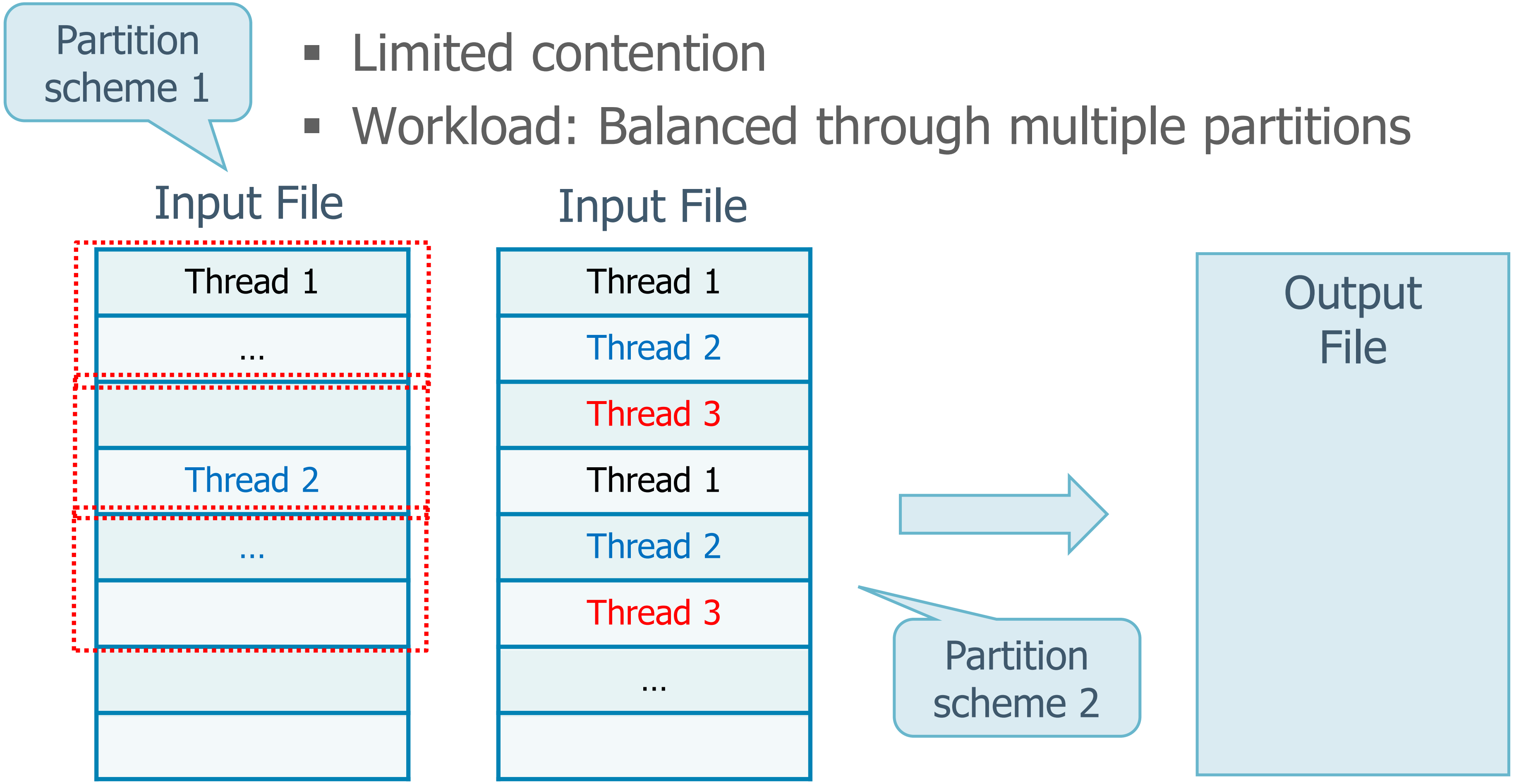
- Solution 2
  - Assign to each thread 1/N of the file (**static** partition)
    - No contention
    - Workload: Efficiency is limited by the slower thread



Solution

- Solution 3
  - As solution 2 but create more partitions than threads as solution 1
    - Limited contention
    - Workload: Balanced through multiple partitions

Each thread gets more sections of the file



Solution

Implementation following solution 1

Includes, variables and prototypes

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <ctype.h>
#include <unistd.h>
#include <sys/types.h>
#include <semaphore.h>
#include <pthread.h>

#define L 100
struct threadData {
  pthread_t threadId;
  int id;
  FILE *fp;
  int line;
  int sum;
};
static void *readFile (void *);
sem_t sem;
```

It must be unique (i.e., it is global or it is passed as parameter to threads through this structure)

Solution

Main
Part 1

```
int main (int argc,char *argv[]) {
  int i, nT, total, line;
  struct threadData *td;
  void *retval;
  FILE *fp;

  nT = atoi (argv[1]);
  td = (struct threadData *) malloc
        (nT * sizeof (struct threadData));
  fp = fopen (argv[2], "r");
  if (td==NULL || fp==NULL) {
    fprintf (stderr, "Error ...\n");
    exit (1);
  }
  sem_init (&sem, 0, 1);
```

I/O Protection-Synch
A mutex is sufficient

Not shared

Init to 1

Solution

Main
Part 2

File pointer common to all the threads

```
  for (i=0; i<nT; i++) {
    td[i].id = i;
    td[i].fp = fp;
    td[i].line = td[i].sum = 0;
    pthread_create (&(td[i].threadId),
      NULL, readFile, (void *) &td[i]);
  }
  total = line = 0;
  for (i=0; i<nT; i++) {
    pthread_join (td[i].threadId, &retval);
    total += td[i].sum;
    line += td[i].line;
  }
  fprintf (stdout, "Total: Sum=%d #Line=%d\n",
    total, line);
  sem_destroy (&sem);
  fclose (fp);
  return (1);
}
```

Solution

Thread function

```
static void *readFile (void *arg){
  int n, retVal;
  struct threadData *td;
  td = (struct threadData *) arg;
  do {
    sem_wait (&sem);
    retVal = fscanf (td->fp, "%d", &n);
    sem_post (&sem);
    if (retVal!=EOF) {
      td->line++;
      td->sum += n;
    }
    sleep (1);  // Delay Threads
  } while (retVal!=EOF);
  fprintf (stdout, "Thread: %d Sum=%d #Line=%d\n",
    td->id, td->sum, td->line);
  pthread_exit ((void *) 1);
}
```

Mutual exclusion for file reading

Exercise 14 (barrier)

- A function f is formed by two distinct sessions A and B, executed only once in sequence

```
... f (...) {
  A
  B
}
```

- The function f is executed by N threads in parallel
  - Each thread can run f at most once
  - N is an integer value, defined but unknown

Exercise 14 (barrier)

- The user would like to synchronize the N threads such that section B is executed by a thread only after all threads have terminated the execution of section A
  - In other words, the N threads must synchronize after running on A and before one of them is running on B
- Write the require C code to implement such a behavior

```
... f (...) {
  A
  B
}
```

Barrier will be formally introduced in Unit 06, Section 07

Solution

```
#define N ...

sem_t m;
sem_t sync;
int nwait = 0;


sem_init(&m, 0, 1);
sem_init(&sync, 0, 0);
```

```
void f(void) {
  A();
  sem_wait(&m);
  nwait++;
  if(nwait==N){
    for(c=0; c<N; c++)
      sem_post (&sync);
  }
  sem_post (&m);
  sem_wait(&sync);
  B();
  sem_wait(&m);
  nwait--;
  sem_post(&m);
}
```

m is the mutual exclusion mutex

sync is the waiting semaphore